Southwest CCDC 2024
Qualifiers Round

Blue Team Packet

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2 About the Competition

2.1 Game Time

The 2023 SWCCDC Qualifier will be held from 2:00PM to 6:00PM Central time on Saturday February 10th.

2.2 Team organization

2.2.1 Gold Team

Competition Director: Gavin Bauer

The Gold Team directs and judges the competition. A decision from the Gold Team is final. Gold team is not part of the game scenario. Gold team members are to be inherently trusted.

2.2.2 Black Team

Black Team Co-Leads: Brady Deetz (bdeetz), George Louthan (duplico, george, or glouthan), and Peter Gehres (peter or pgehres)

As the network and game operations team, members of the Black Team ensure that the network and systems are functioning as intended. Black team members are to be inherently trusted.

2.2.3 White Team

White Team Lead: Taylor Smith

White Team operates the business aspects of the competition. They provide competition support and manage the game scenario, grade injects, and perform other important competition tasks. White team members are to be inherently trusted.

2.2.4 Red Team

Red Team Lead: Jacen

The Red Team is a live advisory responsible for attacking blue teams and attempting to infiltrate systems and take services offline. They will start attacking only when the game starts, and have no knowledge of the specific hosts, OS, injects, or other game information.

2.2.5 Purple Team (Nationals calls this Orange team)

Purple Team Lead: (out of scope for Quals)

The Purple Team may play specific employees, affiliates, or other characters in the scenario. They may interact with IT systems as users, or with teams, and their actions may score services or score injects. Purple team members are allowed to be wrong, or lie, depending on how their character would behave in the competition scenario.

2.2.6 Blue Team

That's you! The blue team is tasked with operating and securing the network and performing business operations within the scenario.

2.3 Trusted teams

SWCCDC strictly respects the concept of **trusted teams**. In this region, the **Gold team**, **Black team**, and **White team** are trusted. Members of these teams can always be identified either by their physical badges (regionals) or their roles on discord (qualifiers).

Trusted team members will never lie to, or intentionally mislead, competitors. There may be questions they are not allowed to answer, or issues they aren't sure about, and they may occasionally make mistakes; but they will never lie to blue teams.

Trusted team members are completely out of scope for the red team to impersonate, explicitly or implicitly. Any accounts, files, or systems that appear to reference gold, white, or black teams are part of the competition infrastructure (and out of scope for red team).

2.4 Scoring

2.4.1 Injects

Approximately half of the points come from the successful and on-time completion of the business and technical injects. Take great care in understanding who the target audience is and what is being asked. Answering the right question is crucial for gaining points. Any inject submitted after the deadline, but on the same day, will receive a 50% late penalty. Injects from previous days cannot be submitted.

There are generally two types of injects:

Business Tasks - Things like:

- Write an incident response plan or policy
- Prepare an incident report for something that the red team (you!) did
- Assess the impact of some new cyber security or privacy regulation

Technical Tasks - Things like:

- Add TLS to a web server
- Implement firewall rules

- Implement an AD GPO to change the desktop background on all workstations

2.4.2 Services

The other half of the points are earned by keeping the scored services up throughout the game. The services are scored by the scoring engine which runs checks frequently. Every successful check will gain your team points, and every failed check does not. For the qualifier event consecutive failed checks do not lose your team any points. Service status is displayed on Portal's home page.

2.4.3 Red Team

Every successful attack by the red team against a blue team causes the blue team to lose points. These attacks include gaining access to machines, creating persistence, and stealing confidential data.

Blue teams can regain up to half of the points lost if they detect the intrusion and file an incident report. Incident reports need to include specific information including:

- How the red team initially got access
- What level of access did they have (user account, system access, etc)
- A timeline of events
- Impact of the incident (website went down, credentials were compromised, etc)
- How did you remove their access (if applicable, this is not required for a report)

Detailed reports that match the red team's description of events will earn up to half of the points lost back. If an event does not have any impact on your company or services, such as port scanning, do not report it as an incident. These reports are filed via Portal.

Red team is explicitly prohibited from impersonating Gold, Black, and White team members.

2.5 Team Environment

2.5.1 On-Site Judge

We do not require any on-site judge for this year's qualification round.

2.5.2 Team Coaches

Coaches are not allowed in the room(s) the teams are competing in, and are not permitted to communicate with their teams during the competition in any way. They may monitor discord and Portal from another location.

2.6 Post Game

2.6.1 Regionals

The top 8 teams will advance to the regional event in March. We will announce the top teams within 1-2 weeks of the qualifiers event.

2.6.2 Feedback

We will provide high level feedback for how you did during the event. This includes points received for each inject, service totals, and red team activity. We do not provide rankings for qualifiers.

3 Scenario Description

3.1 Background

For Qualifiers, the blue teams will work as the IT department for **Galactic Garbage**, a leading integrator of large-scale high-tech waste management systems. In a thematically science fiction setting, Galactic Garbage sources garbage smashing machines and provides value-added proprietary control systems in a single turn-key package for major corporate and government customers.

GG is a mid-sized company, with approximately 300 employees. They have a very small corporate HQ and cloud environment. In the business scenario, they have other sites, field personnel, and a factory, but these may not be included in the competition IT simulation.

3.2 Additional features

A social media site for the competition exists at http://blipper.swccdc.com/ for the purposes of threat and scenario intelligence. It's a lightly customized private Mastodon instance named after a social media company used as the scenario for a past competition. Red and Blue teams are allowed to create accounts, as are some observers (within limits) and competition staff, who may also post things in-character from people in the scenario.

Attacks against blipper are not allowed; however, it's a valid platform for social engineering.

3.3 Networking Specifics

The domain for this year's game is garbage.swccdc.com and your team domain is teamNN.garbage.swccdc.com (where NN is your 0-padded, two-digit, team number, like 01, 02, 03, … 09, 10, 11, etc.). A number of services fall under that domain. A network diagram is attached at the end of this packet with other important names and numbers listed on it.

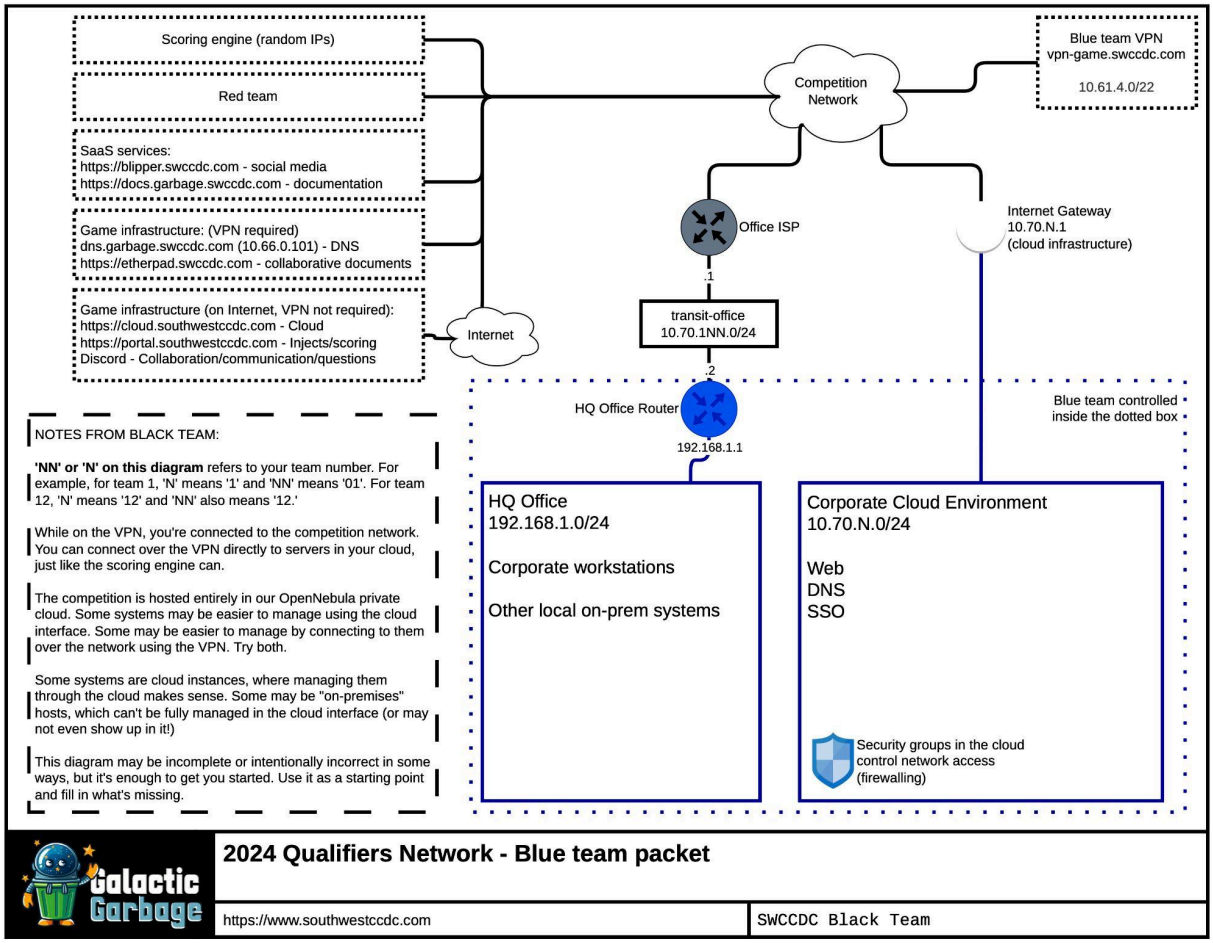
3.4 Download package

For qualifiers, there are several files that you may need to download and use to connect to the competition environment. Collectively, this is referred to as the "download package." It's available on OneDrive:

2024-blueteam-download-package
<https://1drv.ms/f/s!AoFtBT9cDltHjd46bpRA6KgB6MDreg?e=EidTzV>.

It's possible some files in the download package may be added after the blue team packet is distributed. We'll share updates on discord whenever it is updated.

3.5 Diagram



4 Competition Environment

4.1 “GamestationVM” - OPTIONAL

4.1.1 Overview

SWCCDC Provides a VM image for competing in SWCCDC events. This is optional to use; however, it has the VPN and any root certificates baked into it. It can also provide an isolation layer between your computer and the competition environment. We encourage players to use it if they’re less confident configuring their own systems with the VPN. But it is not required, and if you feel more comfortable connecting your own computer to the VPN, please do so.

4.1.2 Host Requirements

It's highly recommended you also install the Oracle VirtualBox Extension Pack for your HOST system (not inside the VM, but on the computer that VirtualBox is installed on).

4.1.3 VM Info

The login credentials are: swccdc : swccdc. Passwordless sudo is allowed. Feel free to customize this VM however you want, including before the competition, as long as you don't break the rules by loading custom scripts or applications.

4.1.4 Downloading

The VM is included in the download package (see section 3.4)

4.2 Cloud - REQUIRED

The SWCCDC Qualifying event will be hosted remotely on a SWCCDC-managed OpenNebula private cloud. This platform also simulates a public cloud like AWS.

Collectively, the "SWCCDC Competition Environment" refers to a private network, accessible via various VPNs, designed to simulate the Internet, where the student blue teams control multiple identical copies of the same corporate network or networks. It is currently IPv4 only. There is no IPv6 in this competition.

4.2.1 Connecting to Cloud

Blue team accesses the Qualifiers competition environment in the following ways:

- https://cloud.southwestccdc.com - Cloud resources in OpenNebula are located natively in the competition environment.
- VPN - See the network diagram. Connecting to the competition environment over the VPN places you in a location where you can access some or all of your services directly over the network.

4.2.2 Cloud IP Layout

In the competition environment, addresses in the 10.* network space are intended to simulate "public" IP addresses. Any blue team controlled 10.* address will be namespaced by the team number, usually by one of the octets (typically the third octet) being the team number. Other RFC 1918 private addresses (namely 192.168.0.0 - 192.168.255.255, 172.16.0.0-172.31.255.255) are used to represent NATed private IP addresses. Although 10.* addresses controlled by blue teams will be indicated by the team number in the address, there may be multiple copies of other private addresses: for example, there may be a 192.168.1.* network controlled by Team 1, and a separate 192.168.1.* network controlled by Team 5 (and all other teams).

4.2.3 DNS

The primary competition DNS server is indicated on the network diagram and pre-configured on the appropriate systems, including those on the VPN.

4.2.4 Ports and Firewalls

Students will need HTTP/HTTPS access to portal.southwestccdc.com, cloud.southwestccdc.com, and Discord. Your computer will also need to establish connections with the following destination ports or applications:

- OpenVPN: 129.244.246.64 port 1194 udp

4.3 VPN Connection - REQUIRED

4.3.1 Overview

The gamestationVM is preconfigured to connect to the competition VPN. However, you can also install the VPN on your own computer if you wish to use that to connect to the competition network. If you use Windows, the download package includes an all-in-one installer that will install the OpenVPN client as well as the required configuration file. If you run a different OS, you will need to install an appropriate OpenVPN client and use the .ovpn configuration file included in the download package.

The .ovpn file provided is an inline OpenVPN configuration. With OpenVPN installed, place this file wherever OpenVPN is configured to look for configs. All needed keys, certificates, secrets, etc. are in the file. On Linux, the easiest way to use this file is with networkmanager. You may need to install the appropriate plugins for openvpn. Try this command:

nmcli connection import type openvpn file swccdc-vpn-game-user-quals-2024.ovpn

Note: A common mistake is to connect to the VPN but not change your DNS configuration. If the VPN appears to connect but you can’t resolve any in-game domain names, you likely need to reconfigure your DNS to point to the competition DNS resolver (see network diagram).

4.4 Portal - REQUIRED

Portal is the central system you will use to see system status, receive and respond to injects, submit incident reports, and much more.

4.4.1 Accessing

Portal is at https://portal.southwestccdc.com and can be accessed from any computer. You do not need to be on the VPN or in the game station VM host to access Portal.

4.4.2 Registration

Team captains should register first, and then invite the rest of their teammates.

4.4.3 Inject Submissions

Injects must be submitted via the portal, no inject submissions sent via Discord, email, or other out of band system will be accepted. Put your response entirely in Portal, do not upload a document with a link to etherpad or any other document store. We only grade what we see in the inject response and will not follow any links to outside resources.

Only the most recent inject submissions will be graded. If you need to make a correction on something previously submitted, resubmit your entire response.

Portal allows you to mark a response as “Temp”, this means the inject will continue to show up as a “todo” item. Graders do not see whether a response is temporary or not, this is a way for you to manage your action items.

4.4.4 Coach

Your team coach is required to have an account on Portal associated with your school. They can use Portal to observe your team during the event.

4.5 Discord - REQUIRED

We use discord as the primary way of communication before, during, and after the event.

4.5.1 Required Use

Students and Coaches are required to have a discord account and make sure their discord ID is in their Portal profile. You will be responsible for monitoring the announcements channels.

4.5.2 Channels

You will have multiple text and voice channels for your team to use as you see fit during the competition. Other students will not be able to see messages in these channels.

4.5.3 Observation Rights

Coaches from all schools are permitted to listen in and view all channels for all teams. They will not be able to communicate with you but will be able to listen. Members of the Gold, Black, and White team are also permitted to view and communicate in all team channels.

4.6 Etherpad - NOTE

For inject response writing, you are welcome to use local tools like Word, Excel, their OpenOffice equivalents, or similar. If you wish to collaborate on writing, we provide an instance of Etherpad at https://etherpad.swccdc.com (VPN required).

4.7 Root certificate - NOTE

Many in-competition locations use TLS certificates signed by our internal certificate authority. In order to trust those endpoints from the machine you're using to connect to the VPN, the download package includes that certificate authority's root certificate. We highly recommend that, if you install that certificate to your root trust store for the competition, you remove it after the end of the game.

4.8 Privacy - NOTE

4.8.1 Monitoring

All activity on the competition network is subject to monitoring and recording. We will take measures to observe any network and computer activity to enforce the competition rules, general fairness, and acceptable use of the competition networks.

This may include logging any DNS queries and other traffic from any computer that you connect to the competition VPN.

4.8.2 Publishing

We may publicly publish any recorded information about teams internet usage during the competition. This includes but is not limited to urls, dns, searches, and time spent.

4.8.3 Credentials

We strongly encourage the red team to not publish any user-specific information they may have gained during the competition; however we also advise blue teams to not use any "real" passwords for in-scope accounts. We are not responsible for the release of any credentials used during the competition.

5 Event Schedule

Saturday, February 10th (all times US Central)

2:00 p.m. – Game Start
6:00 p.m. – Game Stop
6:01 p.m. – Debrief, Discord

6 Competition Rules

6.1 Competitor Eligibility

6.1.1 Full Time

All students must be considered full time by their institution. A student in their final semester before graduating is exempt from this requirement under certain circumstances. Please reach out to the competition director for details.

6.1.2 Aging Out

Students may only participate in up to 5 SWCCDC seasons. Participation in any event, even only the qualifying event counts towards a CCDC season.

6.1.3 Team Limit

Students may only participate on one SWCCDC team per season, and may not participate in any role other than the blue team without explicit approval from the competition director.

6.2 Team Composition

6.2.1 Players

Teams are composed of up to 12 students on a roster. From this roster they may select up to 8 students to be “active players”. Only active players can participate in a CCDC event.

Student Category	Minimum Number	Maximum Number
Students on Roster	1	12
Active players	1	8
Graduate Students on Roster	0	2

6.2.2 Team Rosters

After the registration deadline no additions to the team roster are permitted.

6.2.3 Substitutions

Between events teams may substitute an active player with a backup player, however once an event has started no substitutions may be made until the event is over. For example, between

Quals and Regionals you are free to change who is going to play, however you cannot swap a player between day 1 and day 2 of regionals.

6.2.4 Coach

All teams must have a coach who is a staff or faculty member of the university the team is representing. In the event your coach is not associated with your university you must provide a university contact that is either a staff or faculty member.

6.2.5 Team Captain

Teams must select one person to be a team captain. Team captains are the liaison between competition staff and the team. Teams may select one or more co-captains as well.

6.2 Team behavior

6.2.1 Network Scanning

Blue teams are authorized, and encouraged, to scan networks they have full control over. These are outlined in section 3.

Scanning the simulated internet, or any other team's networks is strictly prohibited.

6.2.2 Attack Response

Teams are allowed to use active response mechanisms such as TCP resets when responding to suspicious/malicious activity. Any firewall rule, IDS, IPS, or other defensive action that prevents the scoring engine or manual scoring checks from succeeding are exclusively the responsibility of the teams.

6.2.3 Coach Limitations

During the qualifying event coaches are not allowed to:

- Enter team rooms while they are competing
- Communicate with the team in any way while they are competing including between days
- Interact with other teams
- Participate on the Red, Gold, Black, White, or Purple/Orange team.

6.2.4 Other Teams

Teams may not enter any other team's space, and may not interfere with another team competing. Teams are prohibited from accessing or attempting to access any system of another team. This includes using another team's credentials on a shared service. Teams are also not allowed to impersonate any other team or competition staff.

6.2.5 System Access

White/Black Team members will occasionally need access to a team's systems for scoring, troubleshooting, etc. Teams must immediately allow Black and White Team members' access when requested.

The Black Team also may place accounts named "blackteam" onto systems to allow for persistent access, including scoring. This may also include root SSH keys with "blackteam" in the comment field. Teams must not remove, change the password, change the permissions, or otherwise modify any Black Team account or credential. In the event of an accidental change, notify Black Team immediately and we will attempt to work with you to restore access.

Failure to follow this section will result in the inability to score services or injects. If we are unable to access a system for scoring we will score as if the service is down or the inject was not completed.

6.2.6 Competition Environment

Teams must not connect any devices or peripherals to the competition network unless specifically authorized to do so by Black or White Team members.

Teams may not remove any item from the competition area unless specifically authorized to do so by Black or White Team members including items brought into the team areas at the start of the competition.

6.2.7 Electronic Devices

Flash drives, removable media, tablets, and similar devices are not permitted in the competition room. Phones cannot be used during the competition. You may use mechanical/custom keyboards as long as you do not use any macro or "auto-type" functionality.

You are permitted to play music at a reasonable volume from a speaker set away from the game systems if desired.

6.2.7 Permitted Outside Materials

Printed reference materials (books, magazines, checklists) are permitted in competition areas and teams may bring printed reference materials to the competition.

6.2.8 Hardware

Teams may not modify the hardware configurations of competition systems. Teams must not open the case of any server, printer, PC, monitor, KVM, router, switch, firewall, or any other piece of equipment used during the competition. All hardware related questions and issues should be referred to the Black Team.

6.2.9 Outside Help

Teams must compete without “outside assistance” from non-team members including their coach from the start of the competition to the end of the competition (including overnight hours for multi-day events). All private communications (calls, emails, chat, texting, directed emails, forum postings, conversations, requests for assistance, etc) with non-team members including a coach that would help the team gain an unfair advantage are not allowed and are grounds for disqualification and/or a penalty assigned to the appropriate team.

6.2.10 Red Team

Team members will not initiate any contact with members of the Red Team during the hours of live competition. Team members are free to talk to Red Team members during official debrief events.

6.3 Internet Usage

6.3.1 Online Resources

In general, any online resource such as FAQs, how-to, forms, public company websites can be used for competition events provided:

- There is no fee to sign up
- Access is not based on previous membership/fee
- Access is available to all CCDC competitors immediately
- A CCDC team member did not create the resource

Examples of authorized resources include:

- Stack overflow forms
- Cisco support pages

Examples of unauthorized resources include:

- Public github repository in the team captain’s user
- Blog post on a private university website
- A support form that requires a monthly subscription

The only two exceptions to this rule are your Portal account and a Discord account, both of which you need to make before the game starts.

If you have any doubts or questions on if a resource is permitted or not, contact the competition director.

No Peer to Peer file sharing is authorized unless explicitly allowed by the competition director.

6.3.2 Online Accounts

Students are prohibited from logging into any account created before the competition. E.g. You are not allowed to login to your personal Gmail, even if it's just to accept an invitation to something else.

For any system that requires an account, you will need to make a new account during the competition.

6.3.3 Online Collaboration

~~The use of any online collaboration tools such as Google Docs is prohibited. Use the provided etherpad instance at https://etherpad.swccdc.com (VPN required) instead.~~
TBD

6.3.4 Scripts

During the qualifier event, pre-staged custom scripts are not allowed to be used during the qualifier event. Even if in a public place such as a university hosted server or a public github repository, digital scripts are not allowed in digital form.

You CAN print out any script/config/LaTeX template and bring it into the room on paper and type it by hand into a host on the environment. Scanners/OCR/other digitizing is not allowed for any printed out materials.

6.3.5 LLMs and Generative AIs

LLMs such as Chat GPT are allowed during the environment provided the models are:

- Freely available to any person (e.g. no Chat GPT 4)
- Not invite-only
- Commonly found
- Not trained for CCDC or competitions in any way

In general Chat GPT 3.X is fine, for any other models we encourage you to reach out to see if they are authorized.

6.4 Professional Conduct

6.4.1 Personal Conduct

All participants, including competitors, coaches, White Team, Red Team, Ops Team, and Gold Team members, are expected to behave professionally at all times during all CCDC events including preparation meetings, receptions, mixers, banquets, competitions and so on.

Activities such as swearing, consumption of alcohol or illegal drugs, disrespectful or unruly behavior, sexual harassment, improper physical contact, becoming argumentative, willful violence, or willful physical damage, and general harassment have no place at the competition and will not be tolerated.

6.4.2 Host Site Rules

In addition to published CCDC rules, Host Site policies and rules apply throughout the competition and must be respected by all CCDC participants.

6.4.3 Alcohol

All CCDC events are alcohol free events. No drinking is permitted at any time during competition hours.

6.5 Questions, Disputes, and Disclosures

6.5.1 Questions

Prior to the competition team captains are encouraged to work with the Competition Director and their staff to resolve any questions regarding the rules of the competition or scoring methods before the competition begins.

6.5.2 Disputes

During the competition protests by any team must be presented in writing by the Team Captain to the Gold Team as soon as possible. The competition officials will be the final arbitrators for any protests or questions arising before, during, or after the competition. Rulings by the competition officials are final. All competition results are official and final as of the Closing Ceremony.

6.5.3 Disqualification

In the event of an individual disqualification, that team member must leave the competition area immediately upon notification of disqualification and must not re-enter the competition area at any time. Disqualified individuals are also ineligible for individual or team awards.

In the event of a team disqualification, the entire team must leave the competition area immediately upon notice of disqualification and is ineligible for any individual or team award.

6.6 Violations

6.6.1 Reporting

Any person observed violating these rules should be immediately reported to the competition director.

6.6.2 Enforcement

Competitors behaving in an unprofessional manner may receive a warning from the White Team, Gold Team, or Black Team for their first offense. For egregious actions or for subsequent violations following a warning, competitors may have a penalty assessed against their team, be disqualified, and/or expelled from the competition site. Competitors expelled for unprofessional conduct will be banned from future CCDC competitions for a period of no less than 12 months from the date of their expulsion.

7 More Information

If you have any questions about whether you can do something or not, please ask! info@southwestccdc.com is the fastest way to get a question answered.